

VZ - FZ series and HS-X/E OLED Display Replacement

Open Hardware Installation Guide

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License:	See the LICENSE files in the project repository

Safety Notice

This document describes installation procedures involving access to internal electronic assemblies of a mains-powered instrument. Installation must be performed only by qualified personnel familiar with electrical safety procedures.

This project is an open hardware display replacement design for repair and restoration purposes. It is not a finished consumer product, not a CE-marked commercial kit, and not intended for standalone operation.

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The documentation is provided as-is and may change without notice.

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1 Safety Notice and Liability Disclaimer

1.1 Electrical Safety Warning

Although the replacement display itself operates at low voltage (+5 V), installation requires opening the synthesizer and accessing the internal power supply unit.

The instrument contains hazardous mains voltages (230 V AC in Europe, 110 V AC in the USA and other regions).

Contact with these voltages can cause severe injury, electric shock, or death.

Capacitors inside the power supply may remain charged even after the unit has been disconnected from the mains.

Caution: Risk of electric shock. Disconnect the instrument from mains power and allow sufficient time for internal capacitors to discharge before opening the unit.

1.2 Qualified Personnel Requirement

All installation and modification work described in this document must be performed by qualified technical personnel with appropriate experience in electronic equipment and mains-powered devices.

This project is not intended for installation by untrained users.

1.3 Liability Disclaimer

This project is provided as-is, without any express or implied warranty or guarantee, to the maximum extent permitted by applicable law.

The installation procedures described herein involve modifications to internal electronics and are performed entirely at the installer's own risk.

The author and project maintainers assume no responsibility or liability for:

- personal injury or death
- damage to the instrument
- damage to other equipment or property
- improper installation or misuse
- modifications performed contrary to this documentation
- regulatory non-compliance of assembled hardware or modified instruments

No guarantee is given for suitability, long-term reliability, regulatory conformity, or compatibility with modified instruments.

By fabricating, assembling, installing, or using hardware based on this project, the user acknowledges these risks and accepts full responsibility for the outcome of the modification.

This document is technical guidance only. It is not legal, regulatory, or professional safety advice.

1.4 General Recommendation

If you are uncertain about any step of the installation process, do not proceed. Consult a qualified service technician.

1.5 Project Status and Intended Use

This project documents an internal OLED display replacement design for compatible Casio and Hohner synthesizer models as described in this document.

The design files, firmware, and documentation are published for repair, restoration, research, and hobbyist use.

This project:

- is not a finished consumer product
- is not offered as a regular retail product or commercial spare part
- is not a CE-marked standalone device or CE-marked kit
- is not intended for independent operation outside the host instrument
- has no functional use outside the designated host instrument
- is intended for technically qualified builders and installers

The author does not provide a commercial installation service and does not assume responsibility for hardware fabricated, assembled, sold, modified, or installed by third parties.

Anyone who manufactures, imports, sells, gives away, distributes, or installs assembled hardware based on this project is responsible for checking and complying with all applicable local safety, EMC, CE, tax, consumer protection, and product liability requirements.

The installer is responsible for ensuring that the modified instrument remains safe and suitable for use after modification.

2 Purpose and Scope

This document describes how to install an OLED display module built according to the VZ - FZ - HS series OLED Display Replacement Open Hardware Project into compatible Casio VZ, Casio FZ and Hohner HS series instruments.

It is intended for qualified technical personnel and experienced hobbyists with practical knowledge of electronics, mains-powered equipment, soldering, ESD precautions, and delicate connectors.

3 Builder and Installer Expectation Checklist

This checklist summarizes normal characteristics of the open hardware display replacement design and its installation. These items are expected behavior and do not indicate a fault.

3.1 Display Characteristics

- The OLED screen size is 2.42 inch and therefore slightly smaller than the original display.
- Small margins around the visible screen area are normal.
- Brightness and contrast differ from the original display technology.
- The display uses modern OLED technology instead of the original display type.

3.2 Installation Characteristics

- The adapter PCB may require slightly more insertion force than the original flex cable.
- The adapter must be fully seated for the display to operate.
- For VZ-1, VZ-10M and HS-2/E units, the replacement module requires a dedicated +5 V supply from the PSU and an upgraded 7805 voltage regulator.
- For FZ-10M, FZ-20M and HS-1/E units, the 7805 voltage regulator does not need to be replaced because these instruments use a different PSU design with sufficient current capability.
- The original VZ-1 metal mounting frame must be reused for the VZ-1 keyboard version.
- Rack versions such as VZ-10M, FZ-10M, FZ-20M, HS-1/E and HS-2/E do not use the VZ-1 metal frame.

3.3 Normal Power-Up Behavior

- The display initializes automatically after the instrument is powered on.
- No user configuration or calibration is required.
- If the display shows a stable image, the installation is considered correct.

3.4 When to Re-check Installation

Re-check the installation if:

- the screen remains blank
- the image flickers
- the instrument behaves unexpectedly after installation

In such cases, first verify connector seating, orientation, and power wiring before assuming a hardware defect.

3.5 Intended Models and Compatibility Notice

This replacement display PCB is designed specifically for the Casio VZ-1, VZ-10M, FZ-10M, FZ-20M and the Hohner HS-1/E and HS-2/E.

The mechanical form factor and connector layout match the original compatible display configuration and fit directly into the corresponding mainboard connector.

Note: The Casio FZ-1 uses a different display form factor and a different mainboard connector. This replacement PCB is not compatible with the FZ-1.

Other models may differ in mechanical design, connector type, or display mounting geometry. It is the responsibility of the installer to confirm compatibility before performing any modification.

Before fabricating, assembling, or installing, users should:

- Verify the model designation of their instrument
- Compare the original display connector with the adapter PCB
- Confirm mechanical clearance and mounting hole alignment

If there is uncertainty regarding compatibility, inspection of the instrument before installation is strongly recommended.

3.6 Required Hardware Assemblies

Depending on the chosen build method, the following hardware assemblies are required. They are listed here as technical installation elements, not as a commercial package or sales offer.

- OLED display replacement module assembly built from the project files
- Adapter PCB built from the project files
- XH wire or equivalent cable to connect the display module with the adapter PCB

3.7 Required Tools

- Phillips screwdriver
- ESD protection, wrist strap or ESD mat recommended
- Flashlight or work light
- Soldering iron and solder

4 Safety and Disclaimer

Caution: Disconnect power before opening the instrument. Unplug the mains cable and wait at least 5 minutes before touching internal electronics.

Note: ESD: The replacement module and instrument electronics can be damaged by static discharge. Proper grounding is strongly recommended.

Note: Do not force anything. If a part does not fit, re-check connector orientation and mechanical alignment.

5 Firmware Information

The display replacement module operates with firmware developed specifically for this hardware platform.

The firmware controls display initialization, timing, and communication with the synthesizer mainboard. No user configuration is required for normal operation after a working firmware version has been flashed.

Firmware source code and, where useful, prebuilt binary release files may be published through the project repository. Firmware may be improved over time to enhance stability, compatibility, or functionality.

Note: Firmware updates are optional and should only be installed if they are explicitly recommended in the release notes. If the display operates normally, no firmware update is required.

For update instructions and the latest information, refer to the project repository.

6 GitHub Repository and Firmware Updates

The project repository contains or may contain the hardware design files, firmware source code, firmware binaries, release information, update notes, bill of materials, PCB manufacturing files, and documentation required to build or service the display replacement design.

https://github.com/DigitalArtDeco/FW_VZ_FZ_HS-XE

Future firmware updates may also be made available through this repository when required. These updates may include improvements related to display timing, compatibility, stability, or additional functionality.

Note: Firmware updates should only be installed if they are explicitly recommended in the release notes. If the display operates normally, no firmware update is required.

Before flashing a firmware file, always verify that the binary is intended for the correct hardware revision and supported instrument model. Using an incorrect firmware file may prevent the display replacement module from operating properly.

Caution: Do not interrupt power during flashing. A failed or interrupted firmware update may leave the display replacement module in a non functional state.

7 Compatibility

- **Tested with:** Casio VZ-1, VZ-10M, FZ-10M, FZ-20M, Hohner HS-1/E and Hohner HS-2/E.
- **7805 regulator upgrade required for:** VZ-1, VZ-10M and HS-2/E.
- **7805 regulator upgrade not required for:** FZ-10M, FZ-20M and HS-1/E.
- **Note:** Instruments modified by previous owners may require additional adjustments.

7.1 Model Specific Installation Overview

The following table summarizes the model specific installation requirements. Always verify the actual PSU and mechanical layout before performing any modification, especially on instruments that may have been modified previously.

Model	7805 regulator upgrade	Dedicated PSU 5 V wiring	Internal metal frame removal	VZ-1 display mounting frame
Casio VZ-1	Required	Required	Not required	Must be reused from the original display
Casio VZ-10M	Required	Required	Required. Internal metal frame behind the display area must be removed.	Not used
Casio FZ-10M	Not required	Not required	Normally no metal frame installed	Not used
Casio FZ-20M	Not required	Not required	Normally no metal frame installed	Not used
Hohner HS-1/E	Not required	Not required	Normally no metal frame installed	Not used
Hohner HS-2/E	Required	Required	Required. Internal metal frame behind the display area must be removed.	Not used

Table 1: Model specific installation requirements for the OLED display replacement.

8 Module Overview and Connectors

8.1 Front side of the replacement PCB

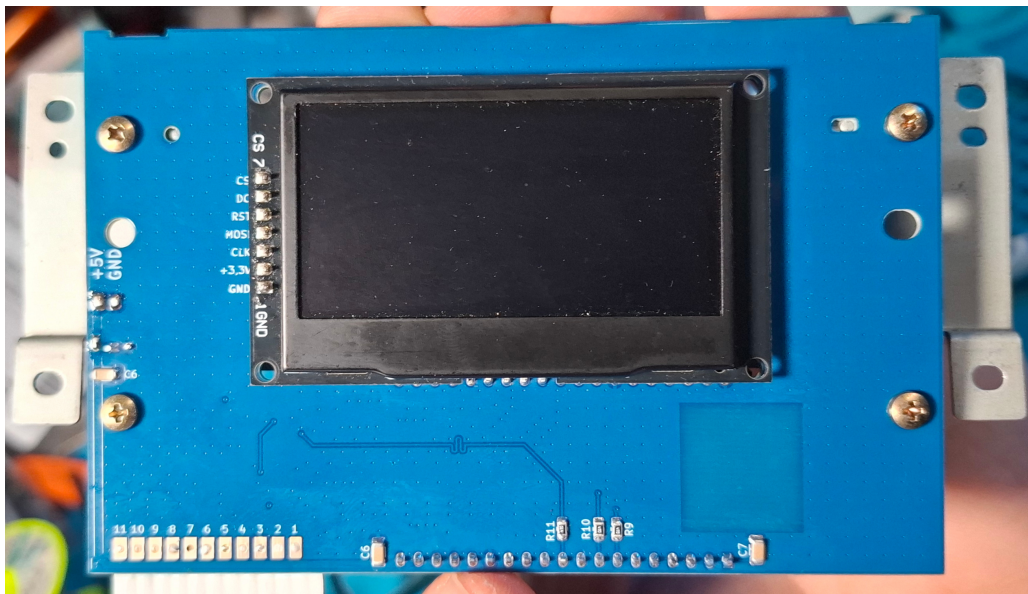


Figure 1: Replacement module, top view.

8.2 Rear side of the replacement PCB

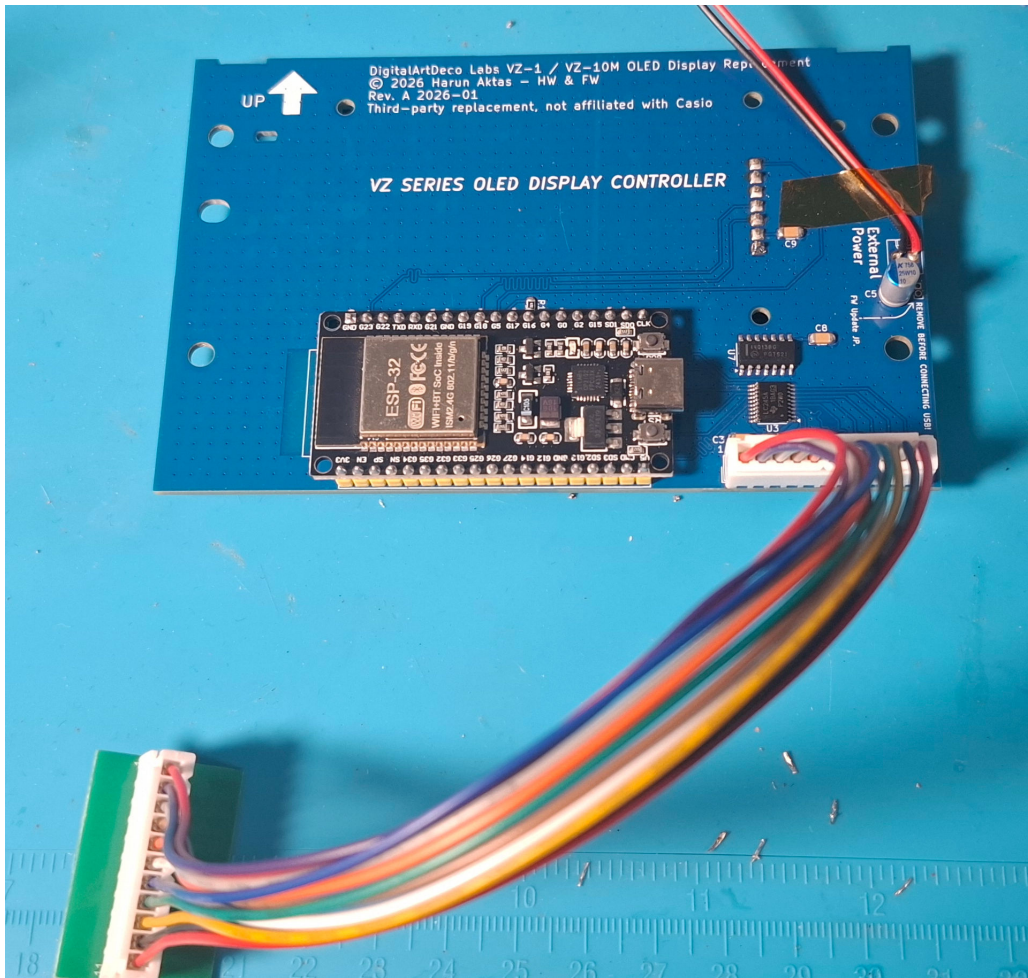


Figure 2: Replacement module with rear view showing the XH connector on the PCB and the connection wire with the adapter PCB.

8.3 Key Areas, Installation Relevant Only

- Display area
- Mounting points
- Mainboard connector location and orientation

9 Preparation

9.1 Work Area

- Clean, well-lit workspace
- Small tray for screws and parts, labelled if needed
- Take photos during disassembly to support correct reassembly

10 Removing the Original Display

Caution: Be careful with fragile connectors and old plastics. Avoid bending or twisting ribbon cables.

1. Locate the original display assembly.
2. Disconnect the display ribbon cable from the motherboard by pulling it out.
3. Remove mounting screws or brackets holding the display.
4. Lift the display assembly out gently.

11 Reusing the Original Display Mounting Frame

11.1 Overview

The original display assembly in the Casio VZ-1 includes a metal mounting frame attached to the display PCB. This frame positions the display correctly inside the instrument and provides the mechanical mounting interface.

For installation of the replacement display in the VZ-1, this metal frame must be removed from the original display PCB and reused on the replacement PCB.

11.2 Removing the Frame from the Original Display

Caution: Handle the original display carefully during removal. Old plastics, connectors, and glass components may be fragile.

1. Locate the four screws securing the metal frame to the original display PCB.
2. Remove all four screws.
3. Carefully separate the metal frame from the original display.
4. Retain the frame and screws for reuse.

11.3 Installing the Frame on the Replacement PCB

1. Align the metal frame with the mounting holes on the replacement PCB.
2. Ensure the orientation matches the original assembly.
3. Reinstall the four screws using the original hardware.
4. Tighten evenly without excessive force.

The replacement PCB includes matching mounting holes and cutouts to accept the original VZ-1 metal frame.

11.4 Model Differences

Note: The metal mounting frame is only used in the Casio VZ-1 keyboard model.

Rack units such as the Casio VZ-10M, FZ-10M, FZ-20M, Hohner HS-1/E and Hohner HS-2/E do not use the original VZ-1 display mounting frame.

This VZ-1 mounting frame must not be confused with the internal metal clearance frame found in some rack units. The internal frame in the VZ-10M and HS-2/E is described separately below and must be removed where present.

The replacement PCB is designed with the required mounting holes and clearances to fit these units directly.

11.5 Important Model Specific Notice

In the Casio VZ-10M and the Hohner HS-2/E, an internal metal frame may be installed behind the original display area.

This metal frame must be removed before installing the replacement display module.

If the frame remains inside the instrument, it may come into contact with the replacement display PCB or its solder pads. This can create an electrical short circuit and may prevent the display or the instrument from operating correctly.

Caution: For Casio VZ-10M and Hohner HS-2/E units, the internal metal frame behind the display area must be removed before installing the replacement display. Leaving this frame in place may cause a short circuit.

Removing this internal metal frame does not affect the normal functionality of the synthesizer. The frame is not required for operation of the replacement display module.

Its removal is recommended to ensure safe clearance, prevent accidental electrical contact, and allow reliable long term operation of the replacement display.

11.6 Removal Procedure

1. Locate the internal metal frame behind the original display area.
2. Carefully remove the frame before fitting the replacement display module.
3. Confirm that no metal parts can touch the replacement display PCB.
4. Check that the display module has sufficient clearance after installation.

11.7 Internal Metal Frame Removal for VZ-10M and HS-2/E



Figure 3: Internal metal frame in the VZ-10M and HS-2/E display area. Remove this frame before installing the replacement display to prevent possible short circuits.

11.8 Illustration for Casio VZ-1

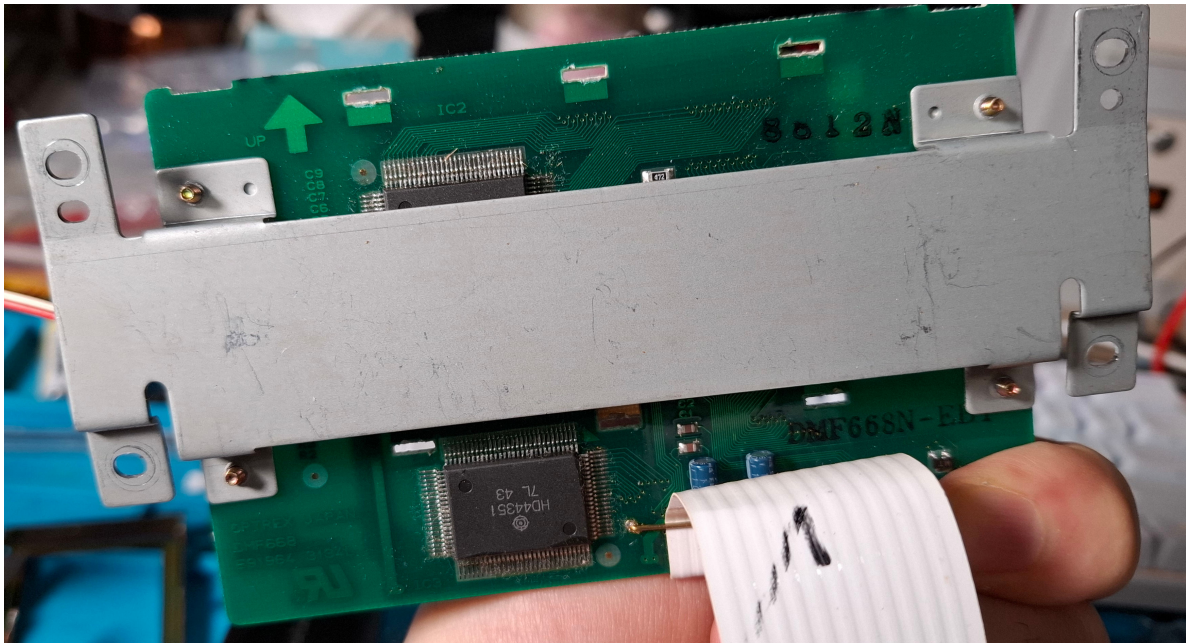


Figure 4: Original VZ-1 metal mounting frame on the old display.



Figure 5: Original metal mounting frame on the new display.

12 Installing the Replacement Display

12.1 Mechanical Installation

1. Place the replacement module in the original display position.
2. Align mounting holes and standoffs.
3. Install screws lightly first, then tighten evenly.

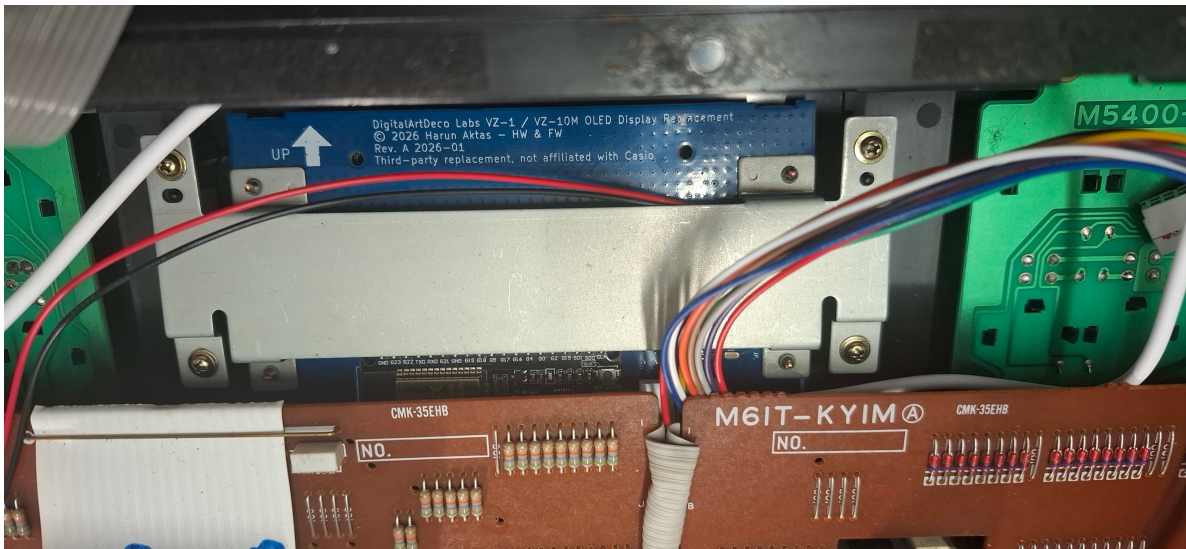


Figure 6: Replacement module installed in the VZ-1.

13 Power Supply Modification, Model Dependent

13.1 Background

The replacement OLED module requires approximately 100 mA additional current at +5 V.

For the Casio VZ-1, Casio VZ-10M and Hohner HS-2/E, the original 5 V power supply design has limited current reserve. In these instruments, the replacement display must be powered directly from the power supply unit and the original 7805 voltage regulator must be replaced with a higher current device.

For the Casio FZ-10M, Casio FZ-20M and Hohner HS-1/E, this regulator upgrade is not required. These instruments use a different PSU design with sufficient current capability for the replacement display.

Note: The 7805 voltage regulator upgrade is required only for VZ-1, VZ-10M and HS-2/E. It is not required for FZ-10M, FZ-20M or HS-1/E.

14 5 V Regulator Upgrade and Thermal Considerations

14.1 Background

The original 5 V supply design differs between the supported instruments.

For the Casio VZ-1, Casio VZ-10M and Hohner HS-2/E, the original 7805 linear voltage regulator does not provide sufficient current reserve for reliable long-term operation with the replacement OLED display and integrated microcontroller.

Although the instrument may power up and operate initially, the original regulator can reach elevated temperatures during extended operation. Thermal stress may result in increased noise coupling into the audio path, voltage instability, or unreliable display operation.

For these models, the original 7805 voltage regulator must be replaced with a higher current 5 V regulator.

This power supply update is not necessary for the Casio FZ-10M, Casio FZ-20M or Hohner HS-1/E. These models use a different PSU design and provide sufficient current capability for the replacement display under normal operating conditions.

Note: Regulator replacement required: VZ-1, VZ-10M and HS-2/E.

Note: Regulator replacement not required: FZ-10M, FZ-20M and HS-1/E.

14.2 Regulator Requirements

For VZ-1, VZ-10M and HS-2/E installations, the replacement regulator must meet the following criteria:

- Fixed 5 V output
- Minimum 1.5 A continuous current rating
- TO-220 package, pin-compatible with the 7805 footprint
- Proper heatsink mounting capability

14.3 Thermal Considerations

Linear regulators dissipate power according to:

$$P_{loss} = (V_{in} - 5 V) \times I_{load}$$

Depending on the PSU input voltage and total system current, significant heat may be generated. The replacement regulator must:

- Be firmly mounted to the existing heatsink
- Use proper thermal compound if originally present
- Have adequate airflow inside the enclosure

14.4 Replacement Procedure

This procedure applies only to VZ-1, VZ-10M and HS-2/E units.

1. Disconnect mains power and allow capacitors to discharge.
2. Locate the original 7805 regulator on the PSU board.
3. Desolder and remove the regulator.
4. Install the higher-current 5 V regulator.
5. Mount securely to the heatsink.
6. Inspect solder joints carefully.

14.5 Model Notes

The 7805 voltage regulator replacement is required for the following models:

- Casio VZ-1
- Casio VZ-10M
- Hohner HS-2/E

The 7805 voltage regulator replacement is not required for the following models:

- Casio FZ-10M
- Casio FZ-20M
- Hohner HS-1/E

The reason is the different PSU design used in these instruments. The FZ-10M, FZ-20M and HS-1/E provide sufficient current capability for the replacement display under normal operating conditions.

14.6 Post-Installation Verification

After installation in VZ-1, VZ-10M and HS-2/E units:

- **Before** connecting to the synthesizer, measure the 5 V rail under **no load**.
- Confirm voltage stability.

14.7 Illustration

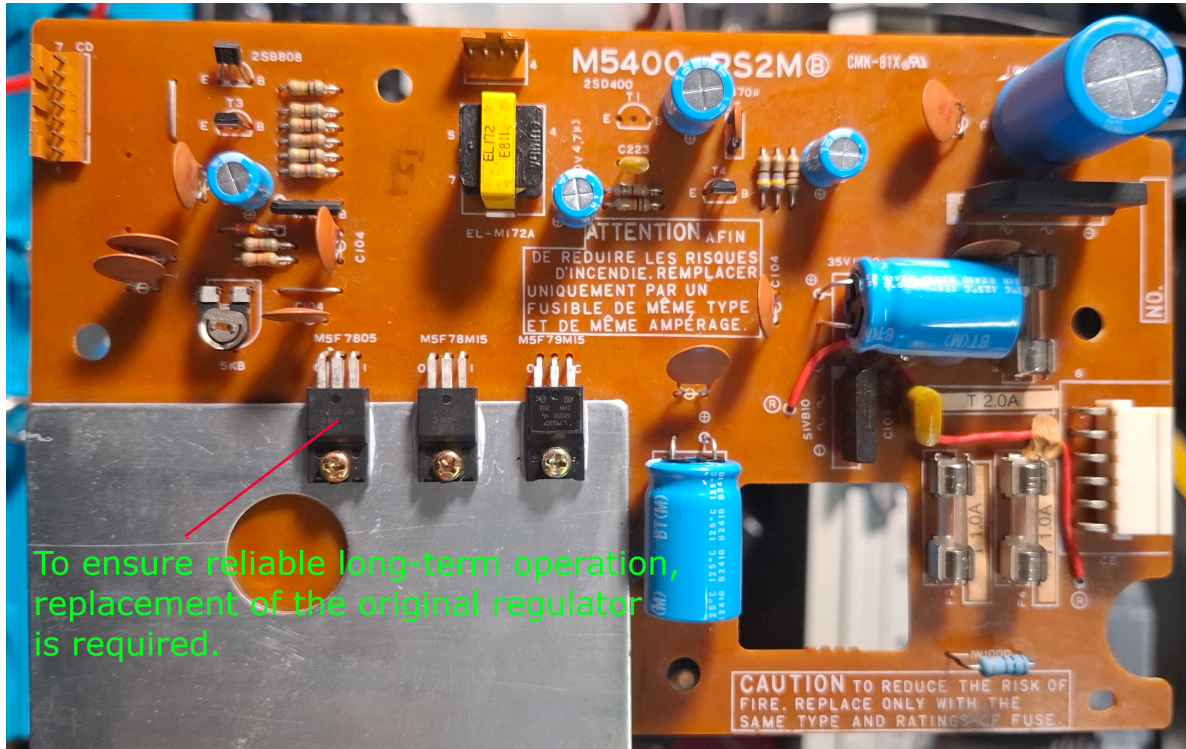


Figure 7: Location of the original 7805 regulator and replacement with higher-current device. Replace with annotated photo.

14.8 Overview

For VZ-1, VZ-10M and HS-2/E installations, the following actions are required:

1. Provide a dedicated +5 V and GND connection from the PSU to the replacement module.
2. Replace the original 7805 voltage regulator with a higher current 5 V regulator.
3. Disable the unused high-voltage inverter circuit where applicable, in order to prevent continuous operation and audible coil noise.

For FZ-10M, FZ-20M and HS-1/E installations, the regulator upgrade is not required.

Caution: Disconnect the instrument from mains power before performing any work on the PSU. Capacitors may remain charged. Wait several minutes before touching the board.

14.9 Adding the +5 V Supply for the Replacement Module

This section applies to VZ-1, VZ-10M and HS-2/E installations.

The power cable used for this installation must be soldered directly to the output of the +5 V linear regulator on the PSU board.

- Red wire → +5 V output of the 7805 regulator
- Black wire → GND

1. Locate the 7805 regulator mounted on the heatsink.
2. Identify the +5 V output pin and a nearby ground point.
3. Solder the red wire to +5 V and the black wire to GND.
4. Route the cable cleanly to the display module and secure it to prevent strain.

This creates a stable and independent supply directly from the PSU and avoids loading sensitive mainboard traces.

Note: For FZ-10M, FZ-20M and HS-1/E units, this additional PSU wiring is normally not required. These models use a different PSU design and provide sufficient current capability through the intended display connection.

14.10 Disabling the High-Voltage Inverter, Transistor T1 / T18

After removing the original display, the high-voltage inverter section may no longer be required. If left active, this circuit may run continuously and can cause audible coil noise.

To disable this unused circuit where applicable:

1. Locate transistor T1 for Casio VZ series on the PSU board. For FZ series locate Transistor T18. See illustrations.
2. Desolder and remove T1 or T18 completely.



Figure 8: Location of transistor T18 for FZ-10M, FZ-20M and Hohner HS-1/E. To disable the high-voltage circuit, remove transistor T18.

Removing T1 or T18 disables the inverter drive stage and prevents unnecessary power consumption and acoustic noise. This modification does not affect normal operation of the replacement display.

Note: The T1 inverter modification applies only where the host instrument uses this inverter circuit. Always verify the PSU layout before removing any component.

14.11 Final Inspection

Before reassembly:

- Check all solder joints for shorts or cold joints
- Ensure wires cannot touch heatsinks or sharp edges
- Verify correct polarity, +5 V and GND
- Secure the cable with tape or tie-downs

Caution: Reversing polarity or shorting the regulator output may permanently damage the display module. Double-check wiring before powering the instrument.

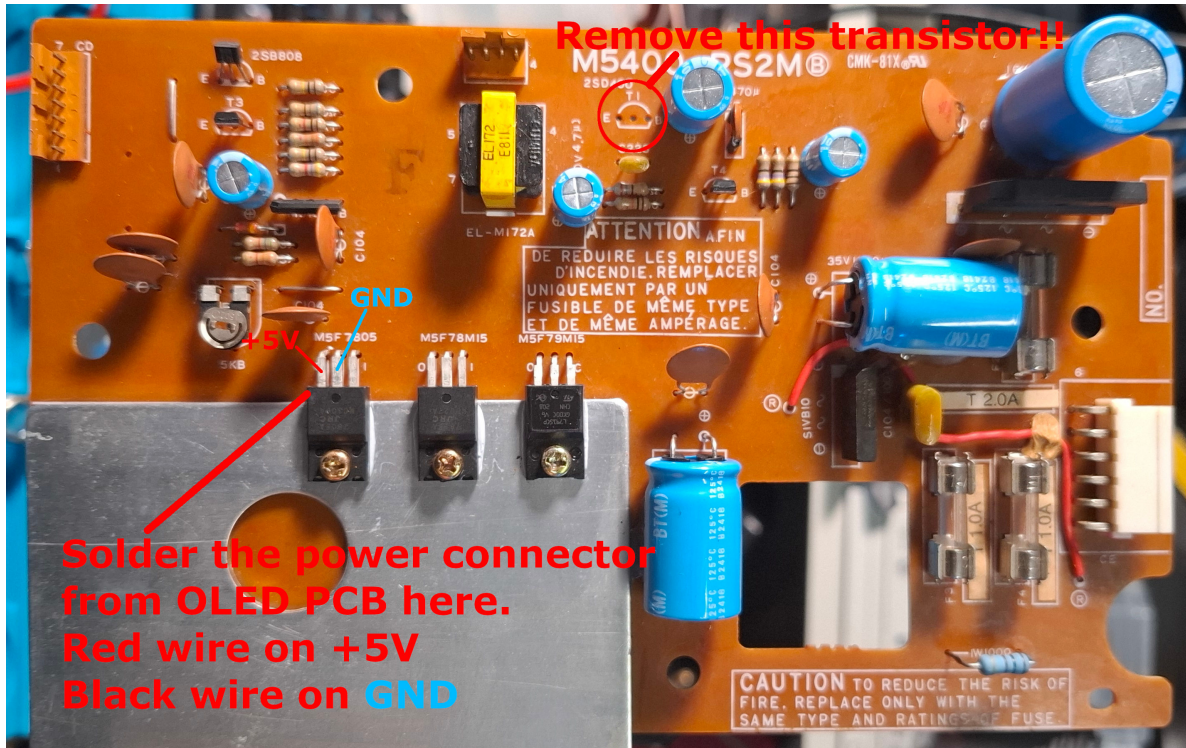


Figure 9: PSU of the Casio VZ series. In this picture the transistor T1 is already removed.

15 Adapter PCB Installation

15.1 Purpose

The adapter PCB provides the electrical and mechanical interface between the replacement display module and the original synthesizer mainboard connector. It replaces the original flexible cable connection with a rigid PCB adapter.

15.2 Mechanical Fit

The adapter PCB is designed to match the original mainboard connector geometry of the synthesizer and fits directly into the existing display socket.

No modifications to the mainboard are required.

Caution: For the Casio FZ-1 a different connector type is used. This adapter PCB is not compatible with the FZ-1. A dedicated version is required.

15.3 PCB Thickness

The adapter PCB thickness is approximately 0.8 mm.

This is slightly thicker than the original factory flex cable. As a result, insertion requires slightly more force than the original part.

This is normal, but care must be taken during installation.

15.4 Insertion Procedure

Caution: Do not bend the mainboard or apply pressure only from one side. Excessive force may damage the connector or PCB traces.

1. Align the adapter PCB with the connector before applying pressure.
2. Ensure the contact pads face the correct direction.
3. Support the mainboard from the rear side with one hand.
4. Press the adapter straight into the connector using even force.
5. Do not tilt or insert at an angle.

Supporting the board from behind prevents mechanical stress and protects the mainboard from flexing or cracking.

15.5 Seating Depth, Critical

The adapter must be fully inserted.

The pads on the adapter PCB must sit flush with the connector.

If the adapter is not completely seated:

- the display may not power on
- the image may flicker or be unstable
- no communication with the display may occur

If the display does not function after installation, first check the seating depth of the adapter before troubleshooting anything else.

15.6 Adapter PCB

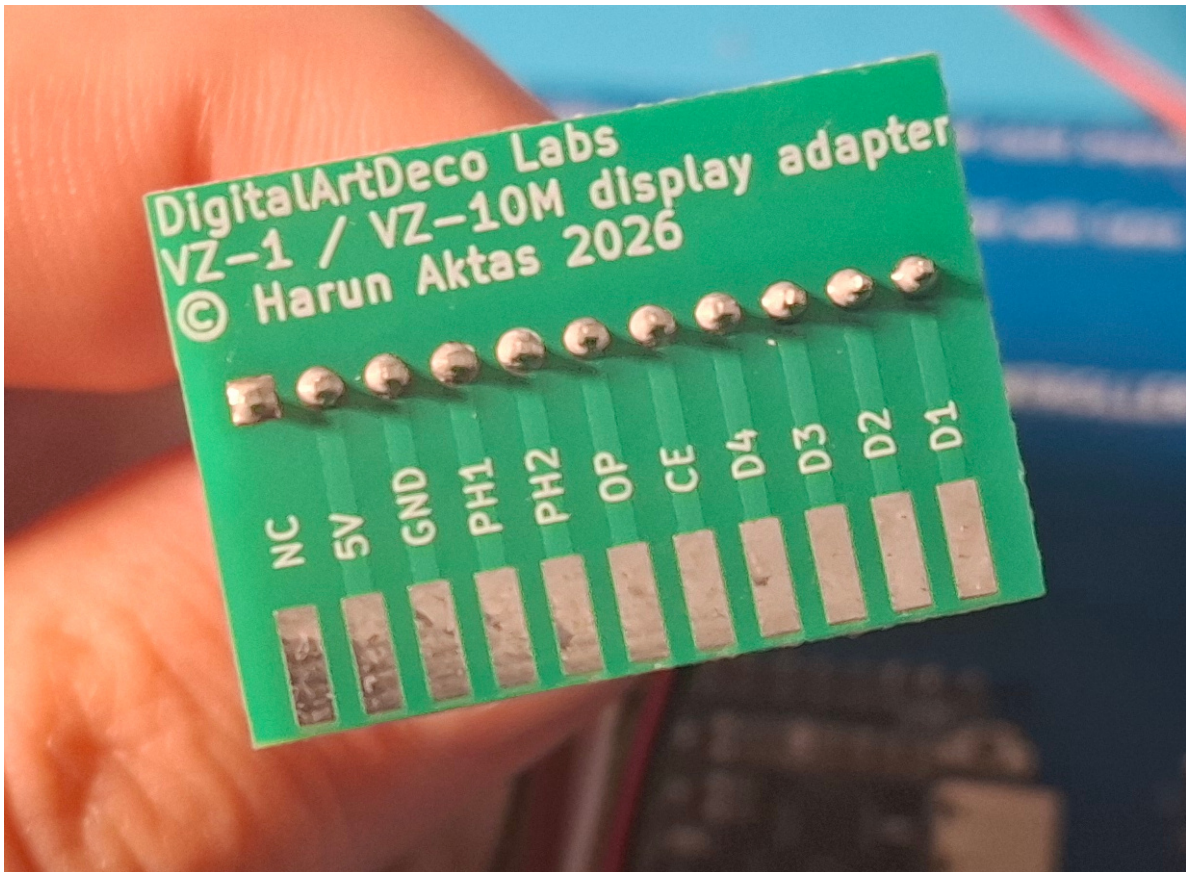


Figure 10: Adapter PCB. Assembly example for the Casio FZ-10M / FZ-20M and Hohner HS-1/E. Please note that the XH connector is installed on the reverse side of the adapter PCB. This is because the Casio FZ-10M / FZ-20M and Hohner HS-1/E mainboard installation differs from that of the Casio VZ-1 and VZ-10M and HS-2/E.

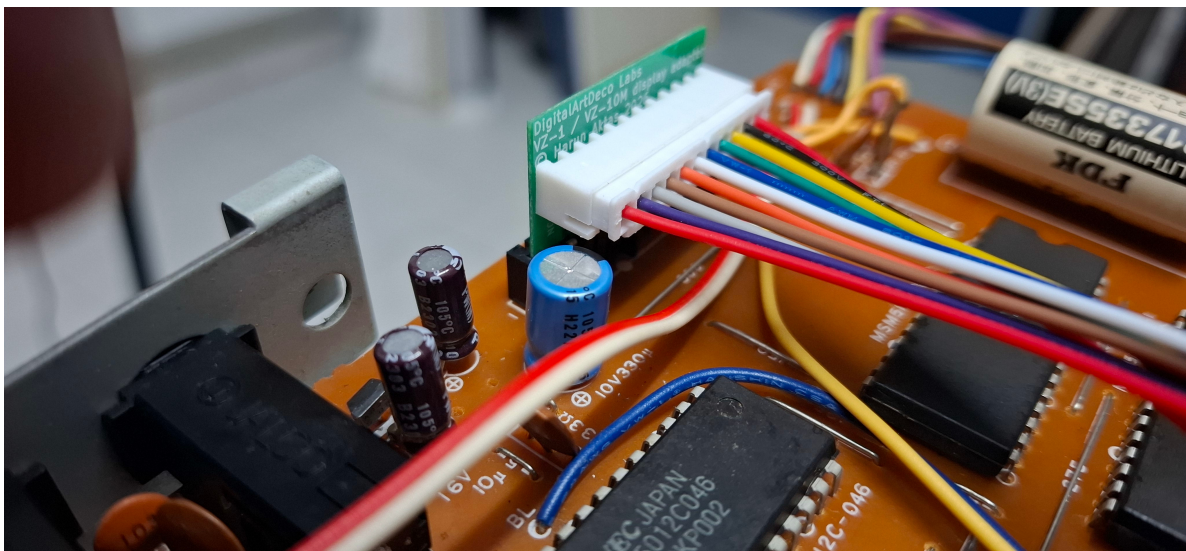


Figure 11: Adapter PCB aligned and fully inserted into the mainboard connector. Pads must be flush. This example shows the Casio version VZ-1 / VZ-10M and Hohner HS-2/E of the installation.

15.7 Connecting the Cable or Cables

Caution: Connector orientation matters. Do not insert a cable at an angle.

1. Verify the connector orientation, pads facing the contact side.
2. Insert the cable fully and evenly.
3. Close the connector latch if present.
4. Perform a gentle pull check to ensure the cable is seated.

15.8 Mainboard Connector Orientation

The original display connector on the synthesizer mainboard is a flex cable socket with electrical contacts located on one internal side only.

For proper operation, the adapter PCB must be inserted so that the contact pads on the adapter face the same direction as the conductive contacts inside the mainboard connector.

1. Inspect the connector on the mainboard and identify the side containing the internal metal contacts.
2. Align the adapter PCB so that its contact pads face this contact side.
3. Insert the adapter straight and fully into the connector.

Note: If the adapter PCB is inserted in the wrong direction, no electrical damage will occur. The adapter is populated with contacts on one side only. In the wrong orientation, the display will simply not function because the signals are not connected.

Correct orientation is therefore required for normal operation even though incorrect insertion does not harm the hardware.

15.9 Orientation Illustration

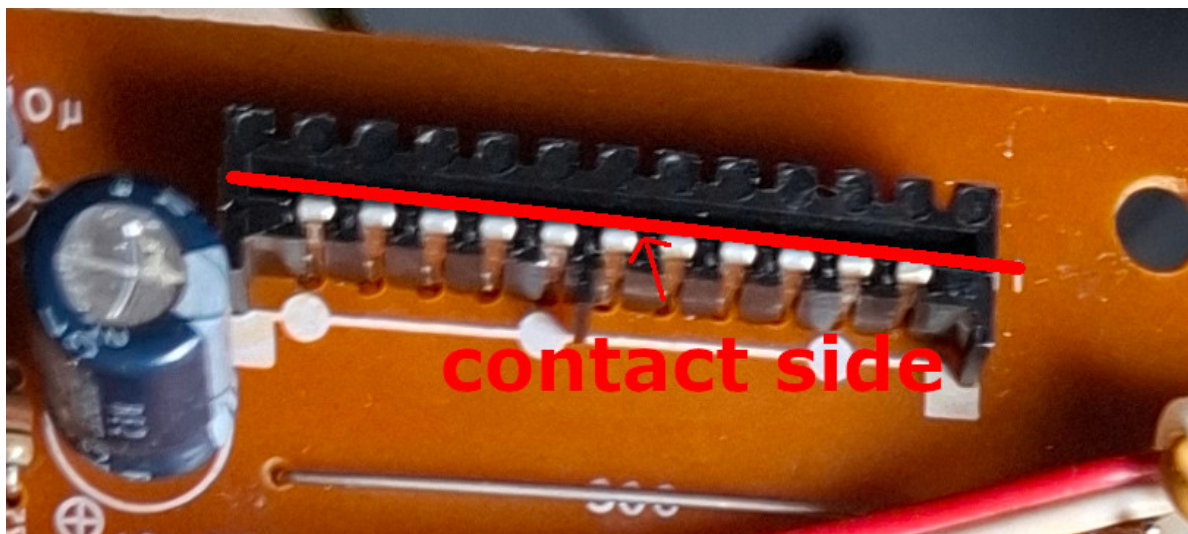


Figure 12: Mainboard connector showing the contact side and the correct adapter PCB orientation.

16 Reassembly

1. Inspect that no cables are pinched.
2. Reinstall covers in reverse order.
3. Reinstall all screws.

17 Power On Check

Caution: If you smell smoke or see unusual behavior, disconnect power immediately.

1. Connect mains power.
2. Power on the instrument.
3. Confirm the display shows a stable image.

17.1 What to Expect

- Normal boot screen behavior.

18 Installed Display, Normal Operation Appearance

18.1 Purpose

This section shows the expected visual appearance of the replacement display when correctly installed and operating inside the instrument.

The images serve as a reference for confirming proper installation and normal functionality.

18.2 What to Expect During Operation

When the display is installed and powered correctly:

- The screen initializes shortly after the instrument is powered on
- Text and graphics appear stable without flickering
- Brightness is uniform across the display surface
- Characters are clearly readable from the normal viewing position

Minor differences in brightness compared to the original display technology are normal and expected.

18.3 Installed Reference View



Figure 13: Replacement display installed in the instrument.

18.4 Installation Confirmation

If the installed display visually matches the reference image and behaves as described above, the installation can be considered successful.

If the display does not initialize, appears unstable, or shows no output, first verify:

- adapter PCB seating depth
- correct connector orientation
- power wiring from the PSU, where required for VZ-1, VZ-10M and HS-2/E

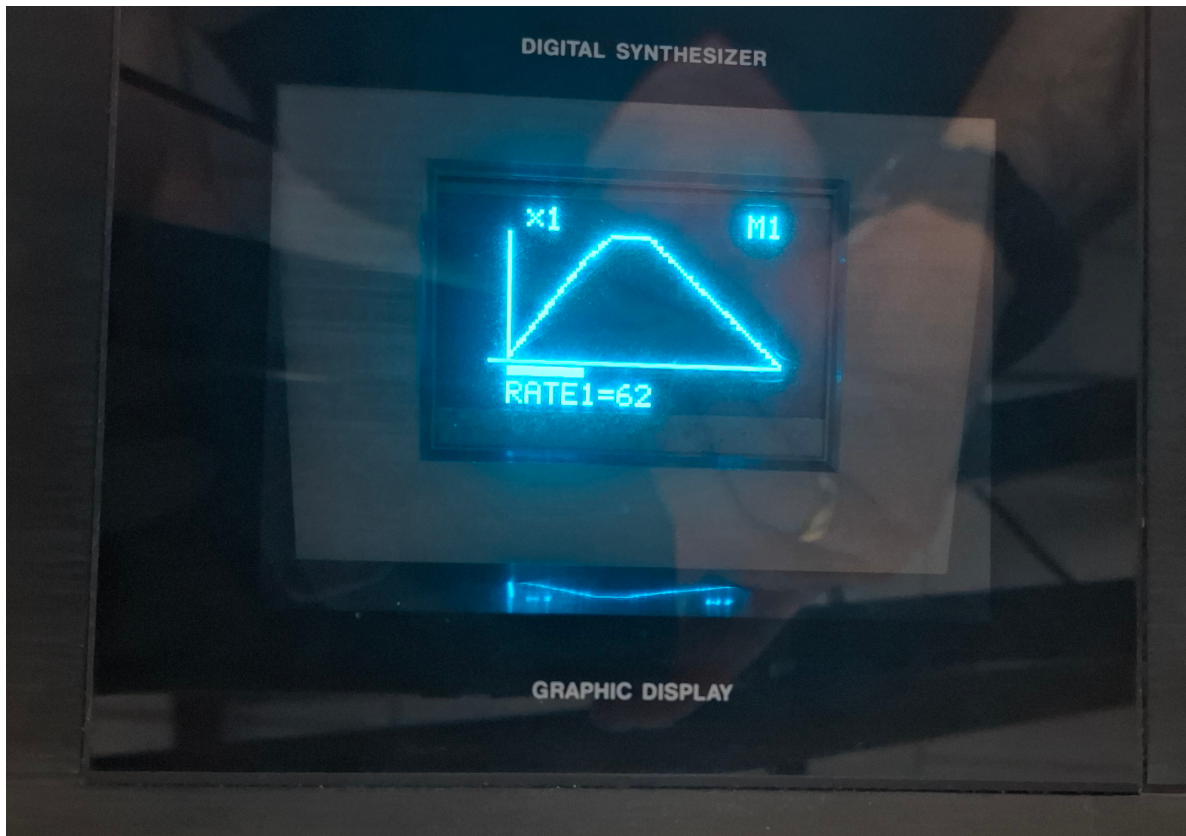


Figure 14: Replacement display installed in the instrument in normal operation mode.

19 Troubleshooting, User Level

Symptom	Action
No display output	Unplug, open again, re-seat the cable and check the connector orientation.
Flickering or unstable image	Check cable seating, make sure the module is mechanically secured, ensure no cable is under stress.
Display works but looks misaligned	Verify the module is seated flush and mounting points are aligned.
Audible noise after installation	For VZ-1, VZ-10M and HS-2/E, verify regulator replacement, PSU wiring and inverter disablement where applicable.

20 Care, Transport, and Maintenance

- Avoid pressure on the display window.
- Keep liquids away from the unit.
- Transport in padded case if possible.

21 Revision History

Version	Date	Author	Changes
1.0	June 23, 2026	<i>Harun Aktas</i>	Initial release with model-dependent 7805 regulator requirements.
1.3.0	June 23, 2026	<i>Harun Aktas</i>	Reworked wording for open source publication, non-commercial project status, liability disclaimer, and CE marking clarification.

22 Regulatory and Safety Information

22.1 General Safety Notice

This document concerns an open hardware display replacement design intended for installation inside an existing mains-powered electronic instrument.

Installation requires access to internal assemblies of the instrument, including areas where hazardous voltages may be present.

Caution: Risk of electric shock. Always disconnect the instrument from the mains supply before opening the enclosure. Do not rely on the power switch alone.

After disconnecting from mains power, allow sufficient time for internal capacitors to discharge before touching the power supply section.

Only qualified technical personnel familiar with electrical safety standards should perform the procedures described in this document.

22.2 Mains Voltage Warning

The synthesizer operates at hazardous mains voltages:

- 230 V AC in Europe and other regions
- 110 V AC in North America and certain other regions

Contact with mains voltage can cause severe injury or death.

The display replacement module itself operates at low voltage (+5 V), however installation requires exposure to mains-powered circuitry.

22.3 Open Source Project Notice

This project is published as open source hardware documentation for repair, restoration, research, and hobbyist use.

The author does not offer this project as a regular commercial product, retail spare part, CE-marked kit, or installation service. Images of assembled boards and installation examples are shown for documentation purposes only and do not constitute an offer to sell.

Publication of design files, firmware, or documentation does not imply that any assembled hardware is certified, approved, CE marked, or suitable for every instrument condition.

22.4 EMI and EMC Considerations

The installation of additional electronic modules may alter the original electromagnetic characteristics of the instrument.

The installer is responsible for ensuring that:

- internal wiring is kept short and properly routed

- ground connections are secure
- no unintended coupling into audio paths occurs
- the modified instrument continues to operate safely and without excessive emissions

Improper installation may increase susceptibility to electromagnetic interference or introduce noise into the audio signal path.

22.5 Regulatory Compliance Notice

The open hardware design files and this installation guide are not a finished commercial electronic product.

The regulatory status of any assembled board or modified instrument may differ from the original factory configuration of the host instrument.

Any modification of mains-powered equipment may affect compliance with:

- EMC regulations
- electrical safety requirements
- regional product safety standards
- local repair, resale, and consumer protection rules

Anyone who fabricates, assembles, imports, sells, gives away, distributes, installs, or otherwise supplies hardware based on this project must determine which legal and regulatory obligations apply to that activity.

The author cannot control third-party fabrication, assembly quality, installation conditions, or later distribution of hardware based on these files.

22.6 CE Marking Status

No CE marking is claimed for this open hardware project, this documentation, the firmware source code, or the design files.

No CE marking should be applied to assembled hardware based on this project unless the responsible person or organization has completed the required conformity assessment, prepared the necessary technical documentation, and accepted responsibility for the resulting hardware and its intended use.

The display replacement design is not intended to be placed into service as a standalone device and has no functional use outside the designated host instrument.

The regulatory conformity of the complete instrument after modification remains the responsibility of the installer, equipment owner, or any party that makes assembled hardware available to others.

22.7 No Warranty and No Guaranteed Support

This project is provided as-is. No warranty, guarantee, or promise of fitness for a particular purpose is given, to the maximum extent permitted by applicable law.

The author may publish updates, corrections, or improvements, but there is no obligation to provide maintenance, installation support, replacement parts, or future firmware updates.

23 Contact

For questions about the open source project and the installation documentation:

- E-mail: akhar9119@gmail.com
- Project repository: https://github.com/DigitalArtDeco/FW_VZ_FZ_HS-XE

- PCB revision documented here: [A](#)

The author does not provide a commercial installation service or guaranteed support response.